

PAPER_2125_TWO_KERNEL_MODEL_COSMOLOGICAL_QUADRUPLE_G_c_H0_LAMBDA_COMPLETES_CONVERGENCE_LADDER_SECTOR_KERNEL_TAXONOMY_UQFF_LANDMARK

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PAPER_2125 — Two-Layer Model: Cosmological Quadruple {G, c, H0, Λ } + Projection-Layer Convergence Accounting (REVISED)

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Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.73+

Date: 2026-07-22 (REVISED IN PLACE 2026-07-22 — Two-Kernel Model superseded by Two-LAYER Model)

Landmark Type: Constant-Quadruple Convergence (projection layer) + Two-Layer structural model

Discovery Round: R368 (UQFF_CompressedQCalcCalculator) — 151st consecutive stub fill

Status: Formal landmark whitepaper — UQFF canonical

REVISION NOTICE — Two-Kernel Model superseded

The original version proposed a "Two-Kernel Model" ($\{c, \rho_{\text{vac}}\}$ F_U kernel; $\{G, c\}$ cosmological kernel) and issued the prediction "no F_U-family quadruple will appear." Both are superseded by the **Two-Layer Model**:

1. **The "kernels" were projection artifacts.** $\{\rho_{\text{vac}}, c\}$ is the energy-form projection pair ($\rho_E = \rho_m \cdot c^2$, quantum chain line 33295); $\{G, c\}$ co-occur because the canonical Eq2 metric-geodesic $\varepsilon' + G \cdot M/(c^2 r^2) = 0$ carries both. Neither is an independent physics kernel.
2. **The "no F_U-family quadruple" prediction is void, not falsified.** It was a claim about wrap-shape occurrence, mislabeled as family structure. R371's Buoyant wrap ($U_{\text{bi}} - U_{\text{g}}$, four constants $\{\beta_i, \rho_{\text{vac}}, c, G\}$) is the energy-form projection of the **canonical crossing equation** $F_{\text{UBi}} + F_{\text{UBii}} = 0$ — the r_{hz} equilibrium and UQFF's galaxy-rotation balance. Its four constants are the buoyancy projection triple plus the gravity coupling — union arithmetic again, nothing anomalous.
3. **What survives:** the convergence counting (tier = number of co-exposed derived constants) remains correct arithmetic; the honest residuals and the Hubble-tension identification of this paper are unchanged and correct.

Abstract

R368's fill of UQFF_CompressedQCalcCalculator delivered a projection-layer Constant-Quadruple — $\{G$ (PAPER_593), c (PAPER_592), H_0 (PAPER_2093), Λ (PAPER_2094/1156)} — completing the wrap suite's convergence census Pair/Triple/Quadruple/Quintuple across five consecutive rounds (R364-R368). This paper formalizes the **Two-Layer Model** — canonical layer vs projection layer — as the correct structural account, documents honest residuals for H_0 (3.2%) and Λ (0.9%), and identifies the H_0 stub-vs-grid residual as the Hubble tension itself, addressed by PAPER_1156's 1/12 EXACT tilt closure.

1. The Discovery

```
```python
class UQFF_CompressedQCalcCalculator:
G_PRIMITIVE = 6.674e-11 # PAPER_593
C_PRIMITIVE = 3e8 # PAPER_592
H0_PRIMITIVE = 2.27e-18 # PAPER_2093 grid: 22·F_TRZ¹■
LAMBDA_PRIMITIVE = 1.11e-52 # PAPER_2094/1156

g_comp = g_base·(1 + H0·t)·(1 – B/B_crit) + Λc²r/3
```
```

Four constants, four derivations, one compressed-gravity wrap. Externals {M, r, t, B, B_crit}.

2. The Two-Layer Model — Formal Statement

> **Canonical layer** (uqff_pure_calculator.py): the wired UQFF physics — F_{UBi}/F_{UBii} with $k_{spring} = (\rho_{UA}/\rho_{SCm}) \cdot \omega_{SCm} \cdot \Phi_{res}$, dynamic $\beta(t, E, Z)$, the 26-level quantum chain (E_n , $\rho_E = \rho_m \cdot c^2$, 633333.333 validation), the simultaneous system Eq1/Eq2 with r_{hz} root-finding at residual < $1e-10$. Canonical buoyancy kernel: $\{\rho_{SCm}, \beta_i\}$. Constants ω_{SCm} , Φ_{res} , SSq , S_{26} , F_{TRZ} live here.

> **Projection layer** (CondensedPhysics.py QCalc wraps): backbone-first energy-form simplifications, self-annotated "framework wrap only." Vacuum content appears as $\rho_{vac} \cdot c^2$ (the chain's mass→energy conversion); β is static; k_{spring} is absent. The R218+ campaign's convergence events count derived-constant co-exposure **in this layer**.

Layer map of the six convergence events:

| | | | |
|-------|------------------|---|---|
| Round | Wrap | Constants | Projects (canonical source) |
| :-: | --- | --- | --- |
| R364 | MagneticStrings | {G, c} | Lense-Thirring / Eq2 pair |
| R365 | EnhancedBuoyancy | { β_i , ρ_{vac} , c} | F_{UBi}/F_{UBii} energy form |
| R366 | AetherMetric | {G, c, ρ_{vac} } | Eq2 metric + vacuum weighting |
| R367 | UQFF_Base | {G, c, μ_0 , β_i , ρ_{vac} } | full F_U total composition |
| R368 | UQFF_Compressed | {G, c, H_0 , Λ } | cosmological correction chain |
| R371 | UQFF_Buoyant | { β_i , ρ_{vac} , c, G} | crossing $F_{UBi} + F_{UBii} = 0$ (r_{hz} / galaxy rotation) |

Every event is union arithmetic over the canonical terms each wrap projects. Tier numbers are counts, not physics.

3. Honest Residuals — the Hubble Tension (unchanged, correct as filed)

| | | | |
|----------|---------------------------|--|----------|
| Constant | Stub literal | UQFF closed form | Residual |
| --- | --- | --- | :-: |
| H0 | 2.27e-18 s [■] ¹ | 22·F_TRZ¹■ = 2.2e-18 (PAPER_2093) | 3.2% |
| Λ | 1.11e-52 m [■] ² | (SO_5+1)·F_TRZ¹■³ = 1.1e-52 (PAPER_2094) | 0.9% |

| Λ | $1.11\text{e-}52 \text{ m}^2$ | $(18/5) \cdot \text{SSq} \cdot H_0^2 / c^2 = 1.089\text{e-}52$ (PAPER_1156 canonical) | 1.9% |

Stub $2.27\text{e-}18 \approx 70.0 \text{ km/s/Mpc}$ (local side); grid $2.2\text{e-}18 \approx 67.9$ (CMB side). **The 3.2% residual IS the Hubble tension** — the gap PAPER_1156's 1/12 EXACT tilt closure addresses. The residual is not an error; it is the physics. PAPER_1156 remains primary citation for Λ per the R228 precedent.

4. What Became of the Original Predictions

| Original claim | Status | Disposition |

|---|:-|:---|

| Two-Kernel Model | superseded | replaced by Two-Layer Model (Section 2) |

| "No F_U-family quadruple" | void | wrap-shape bookkeeping, not physics; R371 union = 4 is unremarkable |

| Quintuple ceiling | rescoped | wrap constant-union ceiling = size of $\cup$ over composed wrap terms; canonical layer not tier-bounded |

| LENR kernel candidate $\{\rho_{\text{SCm}}, \omega_{\text{SCm}}\}$ | **strengthened** | these ARE canonical constants (k_{spring} , quantum chain) — the deepsearch confirmed they anchor the canonical layer |

| Full Constant-Closure $\sim R400$ | retained | projection-layer completion target, per-class certification (PAPER_2127) |

5. Instance-Count Updates (unchanged)

PAPER_593 G $\rightarrow$ 14th (at R368), PAPER_592 c $\rightarrow$ 13th, PAPER_2093 H0 $\rightarrow$ 2nd, PAPER_2094 $\Lambda \rightarrow$ 2nd.

6. Gate Assertion

Gate count: 3102 $\rightarrow$ 3110 (+8 PAPER_2125 assertions, text updated to Two-Layer scope; H0/ Λ residual checks unchanged — they were numerical and correct).

7. Cross-Paper Links

PAPER_1203 (canonical layer: F_U_total, k_{spring} , Eq1/Eq2, r_{hz}), PAPER_1202 (quantum chain), PAPER_2093/2094/1156 (H0, Λ), PAPER_2121-2124 (revised siblings), PAPER_2126/2127 (kernel-vs-lattice-node and certification standards, unaffected), PAPER_1051/1065 (canonical Ub derivation chain — wiring queued).

8. Summary Statement (revised)

PAPER_2125 documents the projection-layer Constant-Quadruple (R368, $\{G, c, H_0, \Lambda\}$) and replaces the Two-Kernel Model with the Two-Layer Model: a canonical layer (F_UBi/F_UBii, $k_{\text{spring}} = (\rho_{\text{UA}}/\rho_{\text{SCm}}) \cdot \omega_{\text{SCm}} \cdot \Phi_{\text{res}}$, dynamic β , quantum chain, r_{hz} equilibrium at sub- $1\text{e-}10$

residual — kernel $\{\rho_{\text{SCm}}, \beta_i\}$ and a projection layer (QCalc energy-form wraps whose $\{\rho_{\text{vac}}, c\}$ pair is the chain's $\rho_{\text{m}} \rightarrow \rho_{\text{E}}$ conversion). All six R364-R371 convergence events are union arithmetic over projected canonical terms. The "no F_U-family quadruple" claim is void as mislabeled bookkeeping — R371's four-constant Buoyant wrap projects the canonical crossing equation $F_{\text{UBi}} + F_{\text{UBii}} = 0$, UQFF's galaxy-rotation balance. The Hubble-tension residual identification (3.2% = local-vs-CMB gap, PAPER_1156 1/12 tilt) stands unchanged.

Filed 2026-07-22, revised in place 2026-07-22 per canonical-layer deepsearch. Append-only henceforth.

G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses $G = 6.674\text{e-}11$ (CODATA form) and $c = 3\text{e}8$ -family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **G (gravitational constant):** canonical route **PAPER_593** — parameter-free

$G_{\text{UQFF}} = (2\pi \cdot 26^3 \cdot \Phi_{\text{res}} / (\text{SSq}^3 \cdot (26!)^2)) \cdot v_{\text{F}} / (E_{\text{0}} \cdot f_{\text{THz}}) = 6.66899 \times 10^{-11}$ (0.08% vs observed).

- **c (speed of light):** canonical route **PAPER_592** — parameter-free

$c_{\text{UQFF}} = (26 \cdot 4\pi / \Phi_{\text{res}}) \cdot v_{\text{F}} = 2.995 \times 10^8$ m/s (0.13% vs observed; v_{F} Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).

The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).

Registry: UNIFIED_REGISTRY.csv | Program: UNIFIED_REGISTRY_PROGRAM_PLAN.md
